6.5 Inequalities in One Triangle

Name _____ Period ____ Date _____

Geometry 6.5 (30-minute plan) · Student sheet

I can…

- list the angles of a triangle in order when I know the side lengths.
- list the sides of a triangle in order when I know the angle measures.
- decide whether three lengths can form a triangle and find the possible lengths of a third side.

Vocabulary · fill in the missing word

Opposite side (of an angle) — the side of the _____ that does not touch the angle

Triangle Longer Side Theorem — the angle _____ the longer side is larger

Triangle Larger Angle Theorem — the side _____ the larger angle is longer

Triangle Inequality Theorem — the sum of any two side _____ of a triangle is greater than the third side length

Part 1 · Notes: fill in the blanks

Triangle Longer Side Theorem: If one side of a triangle is longer than another side, then the angle _____ the longer side is _____.

Triangle Larger Angle Theorem: If one angle is larger than another angle, then the side opposite the larger angle is _____.

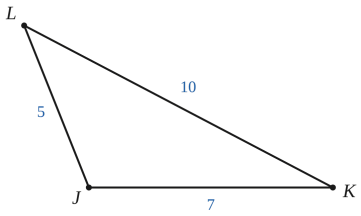
Triangle Inequality Theorem: The sum of the lengths of any two sides of a triangle is _____ than the length of the third side.

Third side: if two sides are a and b, then _____ < third side < _____

Part 2 · Examples (work them with the class)

Example 1

List the angles of △JKL in order from smallest to largest.



Example 2

Can the lengths form a triangle?

a. 4, 9, 12 b. 3, 6, 10 c. 5, 7, 12

Example 3

A triangle has one side of length 12 and another side of length 8. Describe the possible lengths of the third side.

Part 3 · Partner practice

Partner practice

A triangle has sides of length 6 and 15. Describe the possible lengths of the third side.